Goal: make the new PDE code comparable to the old ODE

- The original code, `ode_new`, follows one newborn cohort for 60 months.
- To validate the new age-time PDE code, I first keep the same estimand: one cohort, no warm-up, no continuing births.
- Under this cohort-comparable setting, the PDE code uses the same inputs, same neonatal mortality timing, same IRR-to-VE mapping, and same yearly output columns as the old ODE.
- After this validation, warm-up and continuing births are turned on to represent the full 0–59 month calendar population.

Age-time PDE and cohort-comparable initialization

The age-time population is represented by the transport equation

$$\frac{\partial A_g(a,t)}{\partial t} + \frac{\partial A_g(a,t)}{\partial a} = -\mu_g(a,t)A_g(a,t),$$

where g indexes vaccinated and unvaccinated children.

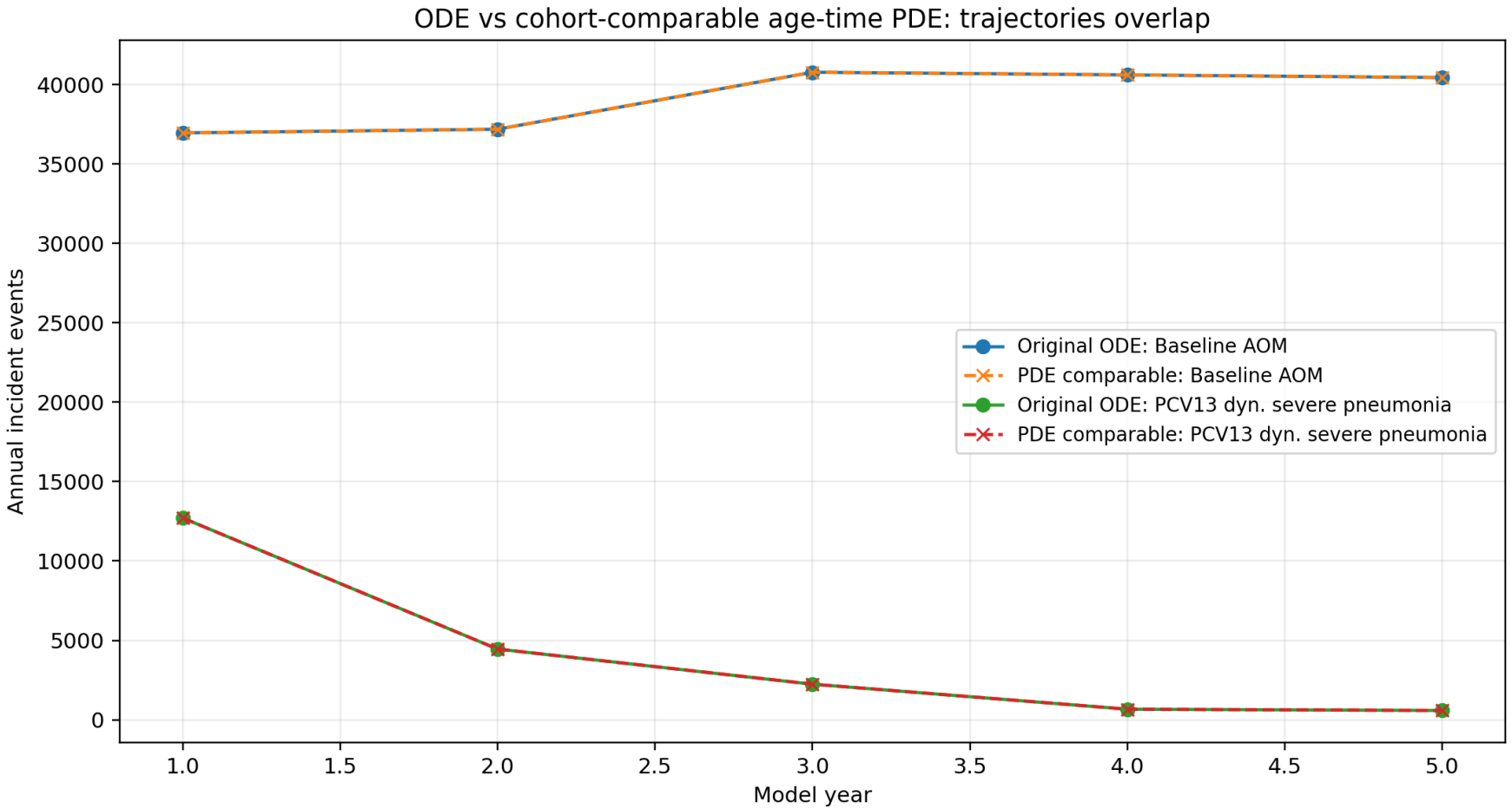
For direct comparison with the original ODE, the validation mode follows one characteristic line:

$$a = t.$$

At recorded month 0, only age 0 is populated. Neonatal mortality has already been applied, all cumulative counts are zero, and all sequela stocks are zero.

ODE/PDE alignment result

- All six yearly CSV outputs match the original ODE after rounding.
- The maximum absolute difference across reported yearly cells is 0.
- The figure shows representative yearly trajectories. The ODE and PDE curves overlap at the output level.



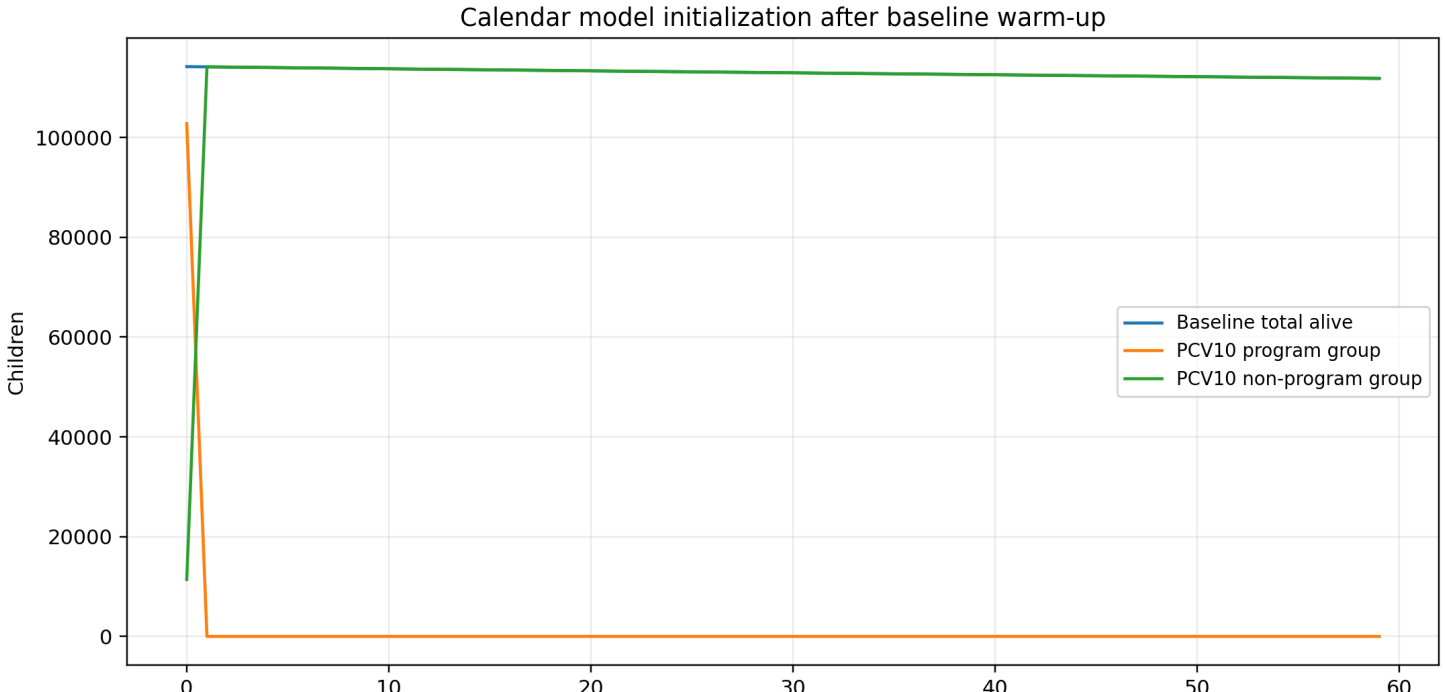
Cohort-comparable PDE five-year outputs

Scenario	Men.	Sev. pneu.	AOM	Deaths	Alive Y5
Baseline	818	46,913	195,947	1,369	1,341,592
PCV10 dyn.	433	23,751	80,212	694	1,342,258
PCV13 dyn.	384	20,654	68,964	604	1,342,347
PCV10 Y5 imm.	213	12,193	50,934	356	1,342,591
PCV13 Y5 imm.	179	10,284	42,962	300	1,342,646
PCV10-SII imm.	181	10,369	43,314	303	1,342,643

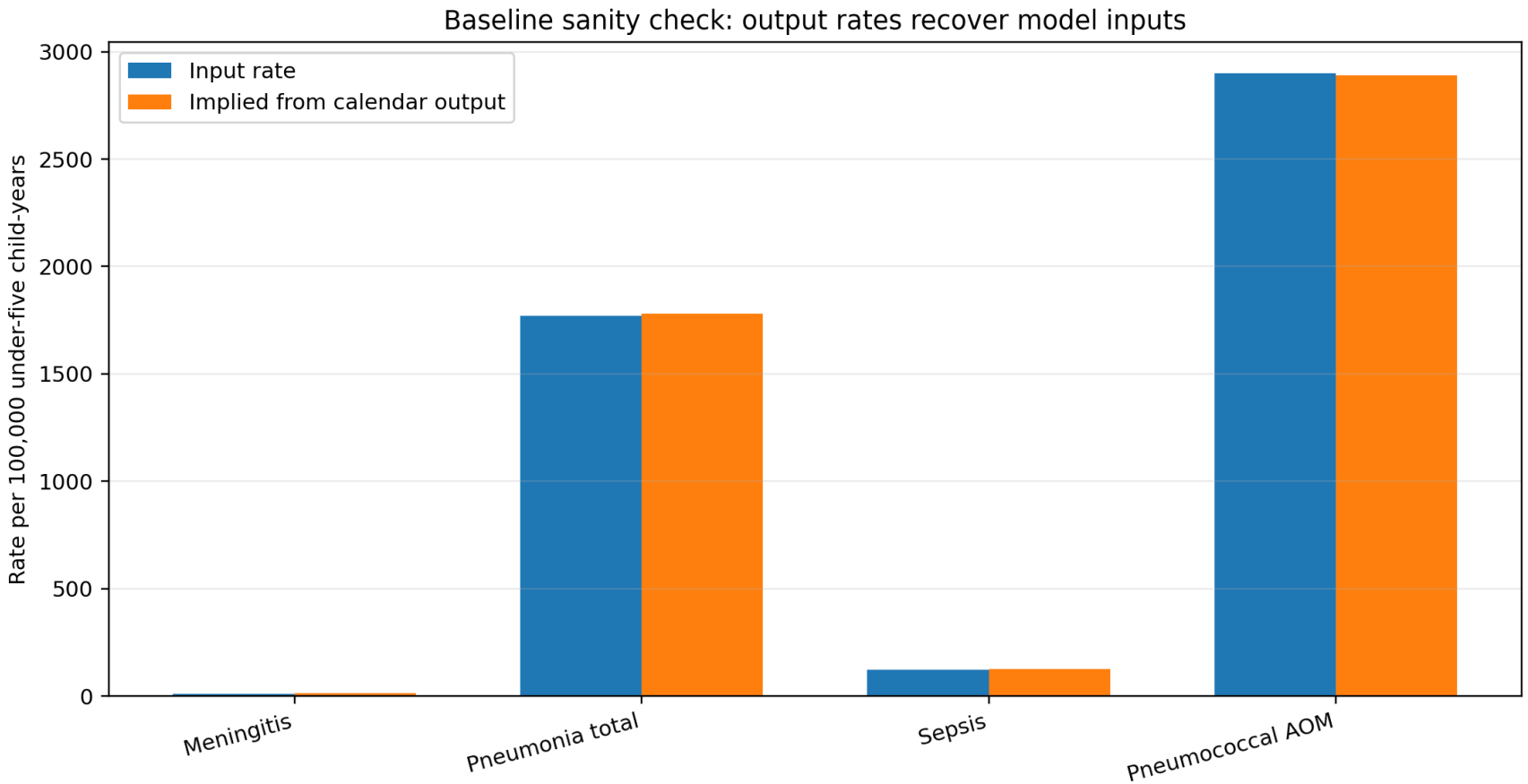
These values reproduce the original birth-cohort ODE outputs under the same assumptions and the same estimand.

Turning on the calendar estimand

- Calendar mode uses a 60-month baseline warm-up to populate ages 0–59 months before the recorded window begins.
- At recorded month 0, the model already contains the existing under-five population.
- Each recorded month adds new births at age 0, applies disease and mortality hazards, advances all children by one month, and removes age 60 from the under-five system.
- This changes the estimand from one cohort life history to full under-five burden in calendar time.



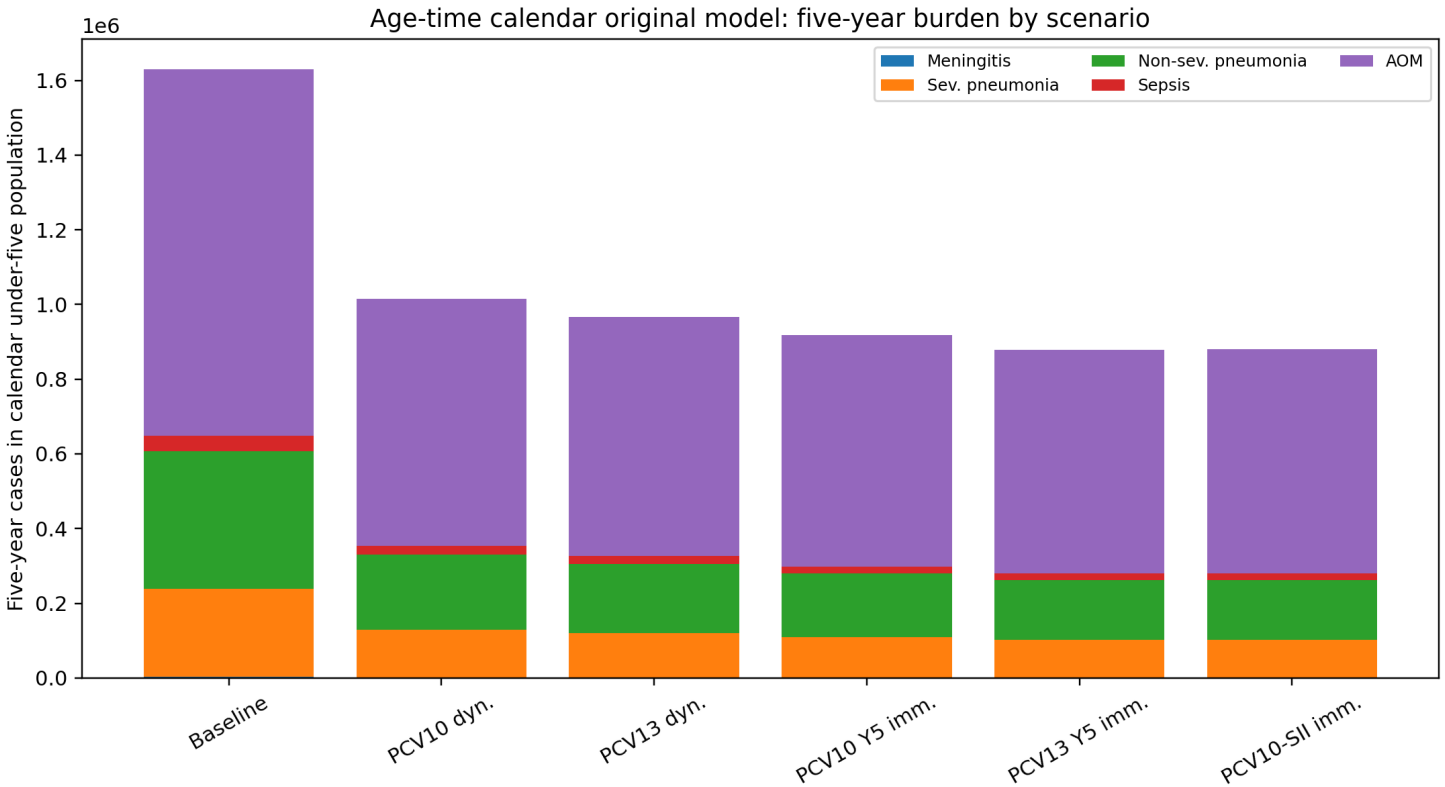
Baseline incidence sanity check



Outcome	Input rate	Implied from output	Relative error (%)
Meningitis	12.00	12.07	+0.59
Pneumonia total	1,771.00	1,779.59	+0.49
Sepsis	124.00	124.60	+0.48
Pneumococcal AOM	2,899.71	2,891.40	-0.29

Calendar-original PDE five-year outputs

Scenario	Men.	Sev. pneu.	AOM	Deaths	Alive Y5
Baseline	4,090	234,570	979,735	6,845	6,776,894
PCV10 dyn.	2,132	127,604	662,789	3,723	6,779,994
PCV13 dyn.	1,954	118,240	639,368	3,448	6,780,265
PCV10 Y5 imm.	1,742	107,917	618,790	3,146	6,780,563
PCV13 Y5 imm.	1,614	100,958	598,949	2,942	6,780,765
PCV10-SII imm.	1,620	101,266	599,827	2,951	6,780,756



Why the calendar outputs are larger

- Original ODE: one newborn cohort accumulated from age 0 to age 59 months.
- Calendar PDE: all children aged 0–59 months are observed during each calendar year.
- Under stable births and mortality, one calendar year of the full under-five population is close to one five-year birth-cohort life history.
- Therefore, the five-year calendar baseline is expected to be approximately five times the original cohort baseline.

What the refined model adds

- The refined model keeps the same age-time calendar population structure.
- Children are split into program-enrolled and not-enrolled groups.
- Direct protection begins after the primary series is active, currently from age 4 months.
- Direct VE is endpoint-specific: IPD, pneumonia, and AOM no longer share one forced VE.

For endpoint d and year y,

$$IRR_{d,y} = H_{d,y}(1 - q_y VE_d).$$

Thus unprotected children use $H_{d,y}$, while directly protected children use $H_{d,y}(1 - VE_d)$.

Recommended comparison structure

1. **Validation:** cohort-comparable PDE vs original ODE. Same assumptions, same estimand, same outputs.
2. **Population change:** calendar-original PDE. Same vaccine mechanism, full under-five calendar estimand.
3. **Mechanism change:** calendar-refined PDE. Same calendar population, refined vaccine mechanism.

